The relationship between mutational spectra and antimicrobial-resistance burden in MRSA

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1 Abstract

1.1 Background

Methicillin-resistant *Staphylococcus aureus* (MRSA) remains a major public-health concern. Although the use of several antibiotic classes has declined in recent years, the evolutionary dynamics of resistance in MRSA remain complex [8]. While AMR is primarily driven by the acquisition of resistance genes, the broader genomic context in which these genes persist is shaped by underlying mutational processes. Mutational spectra, which capture the characteristic patterns of nucleotide substitutions generated by these processes, provide a framework for exploring how environmental and clinical factors shape the evolution of MRSA.

1.2 Methods

I analysed 227 MRSA genomes of CC1 to investigate whether mutational spectra are associated with AMR burden. Resistance genes were identified using AMRFinderPlus, and recombination-filtered core genome alignments were generated with Gubbins. Maximum-likelihood phylogenies were reconstructed with IQTREE and per-isolate mutational spectra were computed from terminal branches using MutTui. Mutational spectra were summarised using principal component analysis, and linear models were used to test whether AMR burden explained variation in global spectra (PC1) or at the substitution-class level (C>A, C>T, T>G). Models included study effects and study-burden interactions to assess whether any associations were consistent across cohorts.

1.3 Results

Across all analyses, AMR burden showed no study-independent association with either global mutational variation or individual substitution classes. Apparent effects observed in simpler models did not persist once study structure was accounted for, and no consistent burden-related signal was detectable across cohorts.

1.4 Conclusions

AMR burden did not show a reproducible, study-independent association with mutational spectra. The conserved mutational landscape of the CC1 lineage likely underlies this lack of detectable burden-related effects.

2 Introduction

2.1 Clinical and epidemiological relevance of MRSA

Methicillin-resistant *Staphylococcus aureus* (MRSA) is a major human opportunistic pathogen and an important cause of healthcare-associated (HA-MRSA) and community-associated (CA-MRSA) infections worldwide. MRSA is defined by the acquisition of the staphylococcal chromosome cassette mec (SCC*mec*), a mobile genetic element (MGE) which carries *mecA* or *mecC* gene and may containt additional various genes, often encoding resistance to other classes of antibiotics [11]. Clinically, MRSA can persist as an asymptomatic colonizer or cause a range of infections, including skin and soft-tissue infections, pneumonia, bacteraemia and sepsis [2].

Epidemiologically, MRSA spreads efficiently in hospitals, nursing homes, and households, with transmission driven by colonized carriers, direct contact, and contaminated surfaces. Distinct HA-MRSA and CA-MRSA lineages differ in their molecular characteristics, virulence, and antimicrobial susceptibility profiles. The global burden of MRSA varies across regions and is shaped by the circulation of specific clonal lineages, making MRSA a continued priority for surveillance, infection control, and responsible antibiotic use [12, 16].

2.2 Characteristics and importance of the CC1 lineage

CC1 has been detected in both community and healthcare settings, and several studies indicate that it can become locally predominant in certain regions or clinical contexts [9, 5, 10]. Importantly, CC1 is frequently described as a colonization-associated lineage: it is commonly recovered from asymptomatic carriers, persists over extended periods, and shows relatively stable prevalence [18]. Such stability contrasts with highly epidemic clones that undergo rapid expansions and contractions. In the UK, CC1 is recognised among CA-MRSA lineages capable of spreading within households, long-term care facilities and general-practice networks, reflecting its ability to move between community and healthcare environments [3]. This combination of wide geographic distribution, sustained circulation, and strong association with colonization makes CC1 a relevant lineage for studying its dynamics and mutational processes.

2.3 Antimicrobial resistance in MRSA

Antimicrobial resistance (AMR) is a defining feature of MRSA and extends beyond β-lactam resistance conferred by the *mec* gene complex. Many MRSA lineages carry additional determinants providing resistance to macrolides, aminoglycosides, tetracyclines and other antibiotic classes, resulting in substantial variation in AMR burden across isolates [17, 1]. These resistance profiles are shaped by antibiotic use patterns and can shift over time, with certain lineages acquiring multidrug-resistant phenotypes while others maintain more stable, moderate resistance repertoires [14]. Recent national surveillance reports indicate a substantial decline in the use of several antibiotic classes in the UK over the past decade [8]. Because AMR burden varies across lineages and over time, and may influence the selective pressures acting on bacterial populations, understanding how resistance relates to mutational processes may provide useful context for interpreting evolutionary patterns in MRSA.

2.4 Mutational spectra as indicators of underlying mutational processes

Mutational spectra summarise the characteristic patterns of nucleotide substitutions accumulated along bacterial lineages and therefore provide insight into the biological processes shaping genome evolution. Different mutational processes leave distinct signatures: oxidative stress tends to increase C>A substitutions [6, 13], defects or variation in DNA repair pathways can alter the balance of transition and transversion events, and selective pressures may influence which mutations persist over time. Because these patterns arise from the combined effects of cellular physiology, environmental exposures and evolutionary constraints, mutational spectra can reveal subtle differences in how bacteria adapt to their ecological or clinical context [15]. In MRSA, where isolates experience heterogeneous environments, including colonisation, infection and antibiotic exposure [7] mutational spectra offer a useful framework for exploring how such conditions may influence evolutionary trajectories.

2.5 Knowledge gap: links between AMR burden and mutational processes in MRSA

Despite extensive work on MRSA evolution and antimicrobial resistance, the relationship between AMR burden and mutational processes remains unresolved. No studies have assessed whether isolates with differing resistance profiles exhibit detectable differences in mutational spectra within a single MRSA lineage.

2.6 Aim of the study

The aim of this study was to examine whether variation in antimicrobial-resistance burden is reflected in mutational spectra within the MRSA CC1 lineage. To address this, I characterised mutational patterns across isolates and evaluated whether resistance-related features show any association with the components of mutational variation.

Table 1: Overview of *S. aureus* CC1 genome datasets included in this study.

Study name	Sampling years	Number of genomes	BioProject	Reference
HPA National	2012-2013	19	PRJEB5707	Toleman et al. 2019
Nursing home	2011-2014	1	PRJEB2655	Harrison et al., 2016
Prospective study	2012-2013	100	PRJEB3174	Coll et al., 2017
ASASM ASARM	2006-2009	21	PRJEB2655 PRJEB2755 PRJEB2791	Saunderson et al., 2014
SCBU	2011	5	PRJEB2737	Harris et al., 2014
BSAC	2001-2010	7	PRJEB5707	Reuter et al., 2016
ASARH	1980	1	PRJEB2655	-
Hepatology	-	2	PRJEB1182	-
HICF	2018-2019	71	PRJEB32286	Blane et al., 2024

3 Methods and Materials

3.1 Genome dataset

I analyzed 227 high-quality *Staphylococcus aureus* CC1 genomes originating from nine independent studies conducted in the United Kingdom. The isolates were collected over multiple sampling periods. All genome assemblies and associated metadata were collected, curated, and kindly made available for this project by Dr Marta Matuszewska. An overview of these studies is presented in Table 1. As a reference genome, I used the RefSeq assembly *S. aureus* SA29-SX (accession: GCF_030533665.1), which served as the coordinate system for downstream analyses.

3.2 Antimicrobial-resistance gene detection

Antimicrobial-resistance (AMR) genes were identified using AMRFinderPlus (version 4.2.7) run on the nucleotide assemblies of all isolates, with the organism parameter set to *Staphylococcus aureus*. All individual reports were merged into a one single csv file for downstream analysis.

3.3 Statistical analysis of AMR class prevalence

To assess whether the prevalence of resistance genes changed over time, AMR class frequencies were compared between the two main datasets representing different sampling periods (Prospective study, 2012; and HICF, 2019). For each AMR class, the proportion of genomes carrying at least one corresponding resistance gene was calculated separately for both datasets. Differences in proportions were tested using a two-proportion Z-test, followed by Benjamini-Hochberg FDR correction.

3.4 Core genome alignment and recombination filtering

Core genome alignment was generated using Parsnp (version 2.1.5), producing an XMFA file that was subsequently converted to FASTA format using Harvesttools (version 1.3). The resulting core genome alignment served as an input for Gubbins (version 3.4.3), which was run with default parameters to identify and remove recombinant regions. The recombinant-filtered alignment was used for phylogenetic reconstruction.

3.5 Phylogenetic reconstruction

Maximum-likelihood tree was calculated using IQTREE (version 2.4.0). ModelFinder Plus (MFP) was used to select the best-fitting substitution model. Branch support was assessed using 1000 ultrafast bootstrap replicates.

3.6 Mutational spectrum inference

Mutational spectra were inferred using MutTui (installed via `pip3 install git+https://github.com/chrisruis/MutTui`). A whole-tree mutational spectrum was computed using the recombination-filtered alignment and VCF files generated by Gubbins, together with the maximum-likelihood phylogeny inferred with IQ-TREE. Based on the whole-tree spectrum, branch-specific spectra were obtained using user-defined label files (e.g., MDR-level, AMR-burden). In addition, MutTui was used to compute per-genome mutational spectra (leaf-level spectra).

3.7 Analysis of mutational spectra

For MDR-level and AMR-burden stratified spectra, cosine similarity matrices were computed and applied to principal component analysis (PCA) using scripts provided with MutTui. Subsequently, PCA was performed on 96-channel leaf-level mutational spectra after filtering genomes by minimum mutation count (n=10). The first principal component (PC1) was modelled using linear regression with three specifications: (i) AMR burden only, (ii) AMR burden with Study (representing sampling period and epidemiological context), and (iii) an interaction term between AMR burden and Study. Finally, the same linear model structures were applied to selected raw mutational signatures (C>A, C>T, T>G).

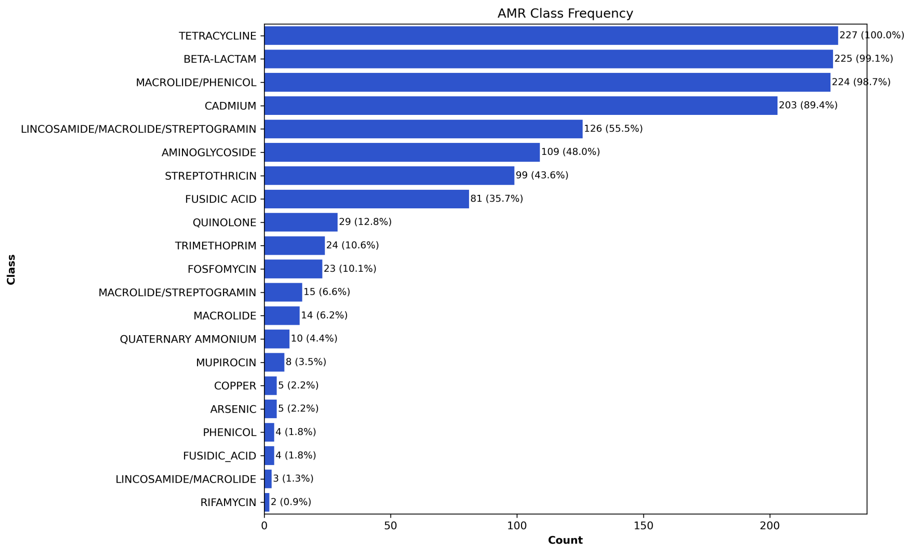

Figure 1: AMR class frequency across MRSA CC1 isolates.

4 Results

4.1 AMR profiles of the MRSA CC1

4.1.1 Distribution of AMR classes

Broad resistance profiles were identified across MRSA CC1 isolates (Figure 1). Several AMR classes were nearly universal: tetracycline (100%), beta-lactam (99.1%) and macrolide/phenicol resistance (98.7%) were present in almost all genomes. Metal-resistance determinants, particularly cadmium, were also widespread (89.4%), suggesting the long-term accumulation of mobile resistance elements in this lineage.

There were also several intermediate-frequency classes. Lincosamide/macrolide/streptogramin resistance occurred in 55.5% of isolates, aminoglycoside resistance in 48.0%, and streptothricin resistance in 43.6%. Several classes were comparatively rare, including trimethoprim (10.6%), fosfomycin (10.1%) and mupirocin (3.5%). This pattern suggests that while CC1 carries a stable core of resistance determinants, accessory AMR elements vary substantially between isolates.

4.1.2 MDR level and AMR burden

MDR levels varied widely across isolates, ranging from 4 to 13 unique AMR classes per genome (Figure 2). Most isolates clustered within MDR levels 6–8, only a small subset exhibited very low or very high MDR levels, suggesting that extreme resistance profiles were uncommon or historically unstable in this lineage.

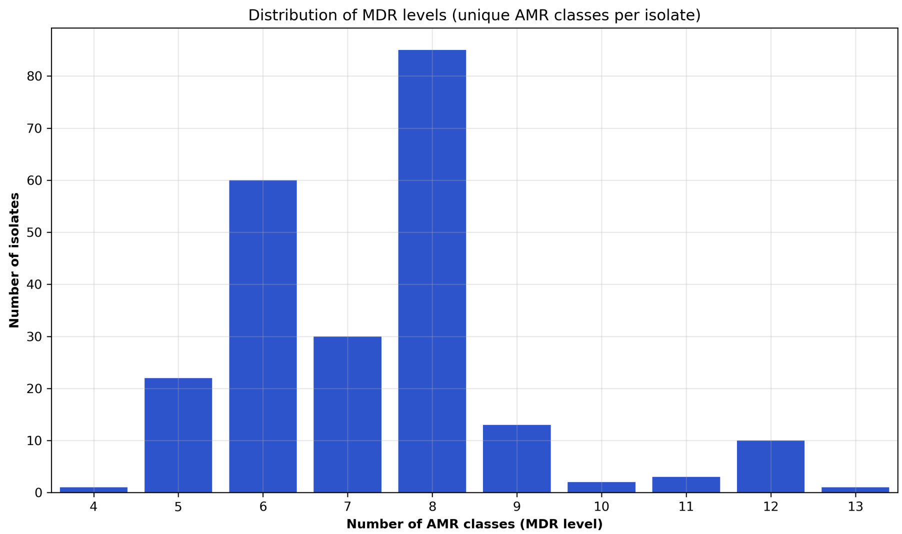

Figure 2: MDR level distribution across MRSA CC1 isolates.

AMR burden also showed broad quantitative variation, ranging from 18 to 39 AMR genes per genome (Figure 3). The observed variation in both MDR levels and AMR burden provides a useful range of resistance phenotypes for exploring potential relationships between AMR profiles and mutational spectra.

4.1.3 Temporal dynamics of resistance to antibiotic classes

Significant temporal shifts were observed in the prevalence of resistance to several antibiotic classes between the two sampling periods (Figure 4. Three classes showed markedly higher resistance in the 2019 study (HICF) compared with the study from 2013: aminoglycosides (0.79 vs 0.36; adj. $p = 2.4 \times 10^{-7}$), streptothricin (0.73 vs 0.33; adj. adj. $p = 8.6 \times 10^{-7}$), and lincosamide/macrolide/streptogramin (0.75 vs 0.53; adj. $p = 0.0108$). Several other antibiotic classes showed identical or nearly identical proportions between the two sampling periods, indicating no detectable temporal change. For a few classes, modest decreases in prevalence were observed over time. This heterogeneity in temporal trends suggests that selective pressures acting on MRSA CC1 have changed over time.

Taken together, the observed variation in MDR level, AMR burden and resistance class frequencies provides sufficient diversity within CC1 to meaningfully explore potential relationships between AMR features and mutational spectra.

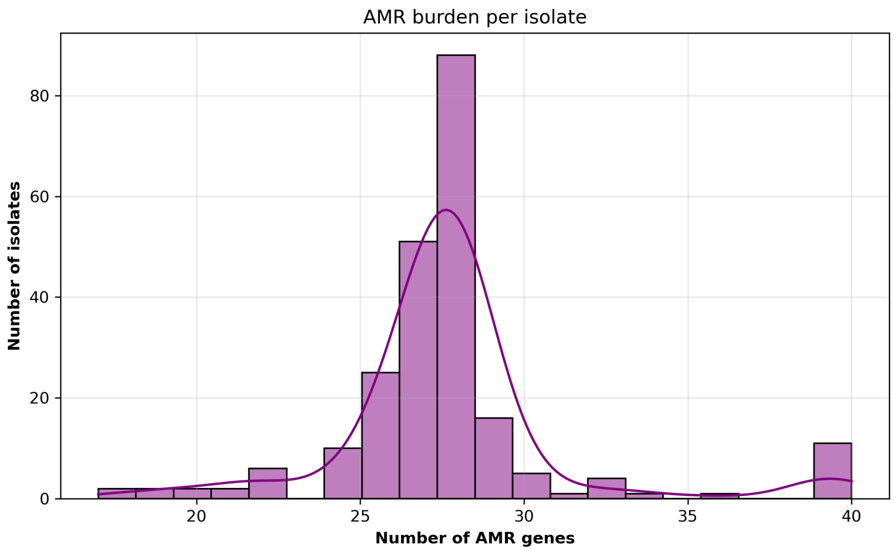

Figure 3: AMR burden across MRSA CC1 isolates.

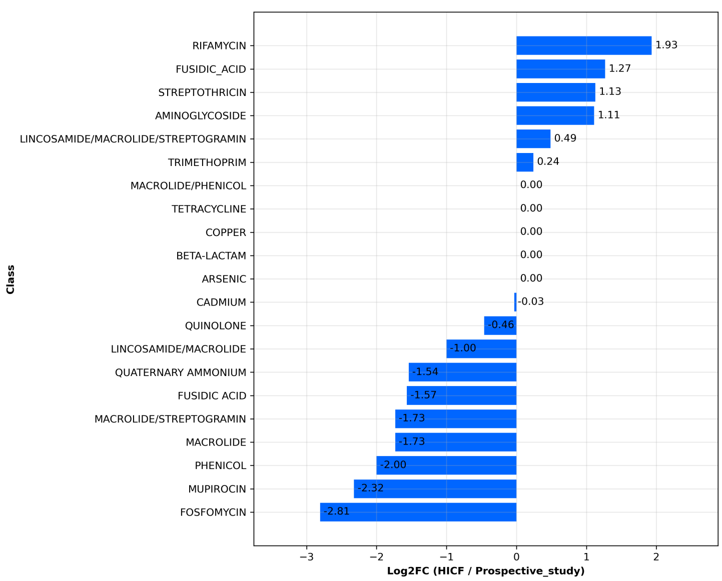

Figure 4: Temporal change of resistance to antibiotic classes.

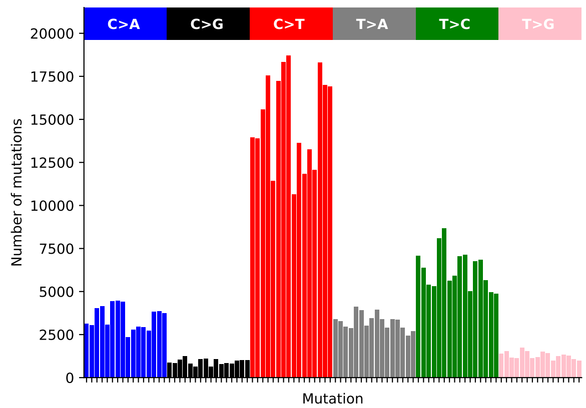

Figure 5: Global spectrum for the whole MRSA CC1 lineage.

4.2 Mutational spectra across MRSA CC1

4.2.1 Global spectrum

The SBS spectrum across MRSA CC1 was strongly dominated by C>T transitions, consistent with spontaneous cytosine deamination, a well-characterised endogenous mutational process in bacteria [4]. Other single-base substitution types occurred at substantially lower frequencies, indicating a relatively homogeneous mutational landscape across the lineage. Overall, the spectrum reflects a stable endogenous mutational background, with no evidence for unusual or stress-induced mutational processes at the population level.

4.2.2 Variation across studies

All studies showed broadly comparable SBS profiles, indicating stable mutational processes across sampling cohorts (Figure 6 . The only deviation was observed in the Hepatology dataset, where three mutation types were absent, most likely due to the very limited number of isolates (n=2).

4.2.3 Variation across STs

Across sequence types, the relative contributions of the six SBS classes were highly consistent. No ST showed a distinctive deviation from the overall CC1 mutational profile, indicating that mutational processes are stable across clonal backgrounds.

Given the overall stability of mutational spectra across studies, sequence types, I next examined whether subtle variation in these spectra corresponds to differences in resistance-related features.

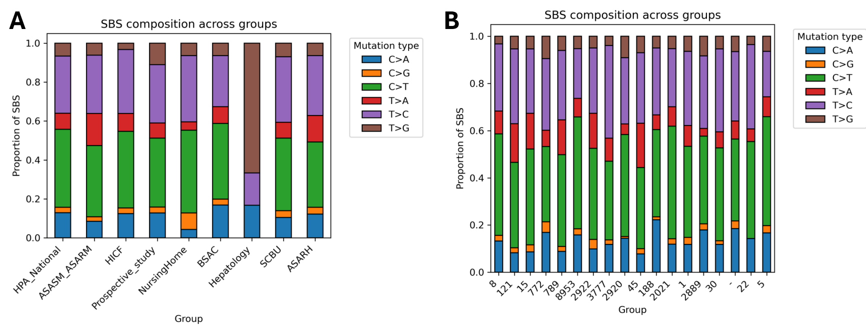

Figure 6: **Distribution of SBS class proportions across MRSA CC1.** (A) SBS composition across studies. (B) SBS composition across sequence types.

4.3 Relationship between AMR features and mutational spectra

4.3.1 Cosine similarity and PCA of mutational spectra in relation to MDR level and AMR burden

PCA based on cosine similarity revealed a single compact cluster of isolates for both MDR level and AMR burden, with only a few clear outliers (Figure 7) . For MDR, the two outlying points corresponded to the extreme categories (MDR 4 and MDR 13), each represented by a single isolate. For AMR burden, the outliers similarly matched the two high-burden categories (31 and 36), which also contained only one isolate each and exhibited very low mutation counts (1 and 4 mutations, respectively, compared with >10 in all other groups). These deviations therefore most likely reflect sparse mutational data rather than genuine differences in mutational spectra, while the main cluster indicates that the vast majority of isolates share highly similar SBS profiles. Given this overall homogeneity, I next examined whether more subtle variation becomes detectable when using the full 96-channel spectra and by modelling PC1 as a function of AMR burden and study-level factors.

4.3.2 Linear models linking PC1 to AMR burden

To evaluate whether AMR burden explains variation in the 96-channel mutational spectra, I fitted three linear models with PC1 as the outcome (Table 2). In the unadjusted model (Model 1), AMR burden showed no association with PC1 ($\beta = -1.0$, p = 0.156). After adjusting for study (Model 2), AMR burden

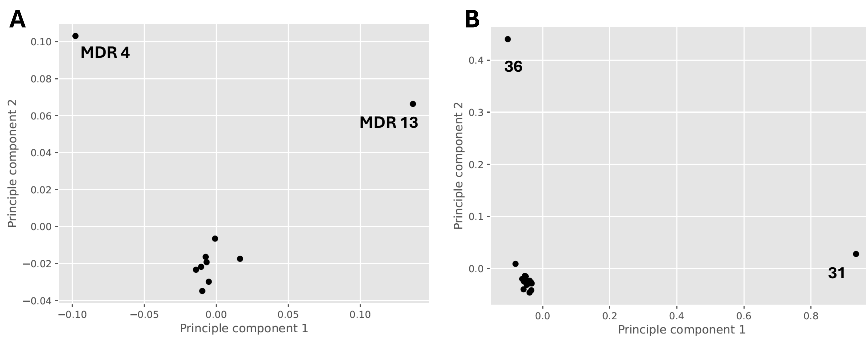

Figure 7: **PCA of cosine similarity of SBS spectra for individual MDR and AMR-burden categories.** (A) PCA for MDR levels. (B) (B) PCA for AMR-burden categories.

Table 2: Linear models testing the association between AMR burden and PC1 from the 96-channel PCA.

Model	Predictors	Effect of AMR burden	Adj. R^2
Model 1	AMR burden	$\beta = -1$, $p = 0.156$	0
Model 2	AMR burden + Study	$\beta = 4.3$, $p = 0$	0.182
Model 3	AMR burden * Study	$\beta = 8.6$, $p = 0.184$	0.229

appeared positively associated with PC1 ($\beta = 4.3$, $p < 0.001$), and the model fit improved (Adj. $\mathbf{R^2} = 0.182$). However, this effect was not stable, when study-burden interactions were included (Model 3), the association disappeared ($\beta =$ 8.6, p = 0.184), and the model showed clear signs of multicollinearity due to the large number of study-specific interaction terms. Overall, none of the models provided evidence for a robust link between AMR burden and the major axis of mutational variation (Figure 8).

To further assess whether specific mutational processes relate to AMR features, I next modelled the proportions of individual SBS classes.

4.3.3 Associations between individual SBS classes and AMR burden

To assess whether specific mutational processes relate to AMR burden, I modelled three specific SBS classes (C>A, C>T and T>G) using the same three model specifications as for PC1. Across all signatures, the unadjusted models (Model 1) showed no evidence of an association with AMR burden (all $p > 0.05$),

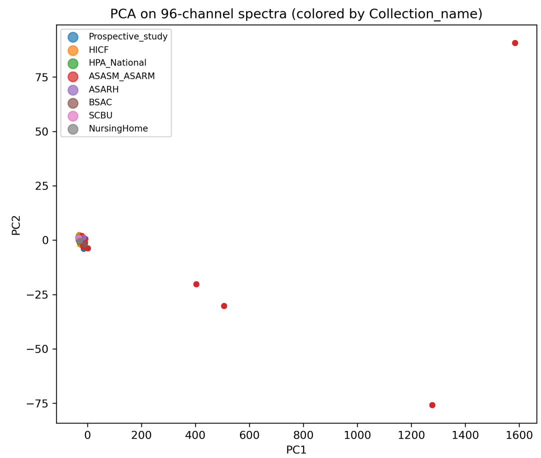

Figure 8: **PCA of 96-channel mutational spectra coloured by study.** Most isolates cluster tightly around the origin, with limited separation between studies. A subset of samples from one study shows greater spread along PC1, indicating modest study-specific variation in the 96-channel spectra.

with small negative effect sizes for C>A (β = –0.64), C>T (β = –3.13) and T>G (β = –0.57). After adjusting for study (Model 2), AMR burden appeared positively associated with all three signatures (C>A: $\beta = 2.56, p < 0.001$; C>T: $b = 10.35$, $p < 0.001$; T>G: $\beta = 1.61$, $p < 0.001$), mirroring the pattern observed for PC1. However, these associations did not persist once study-burden interactions were included (Model 3).In the interaction models, none of the signatures showed a stable or study-independent association with AMR burden. Although the main effect reached nominal significance for C>A ($p = 0.010$),, this reflected only the reference study and was contradicted by large, inconsistent study-specific interaction terms. For C>T, T>G the main effects were clearly non-significant ($p = 0.226$). Across all three signatures, the interaction models exhibited substantial multicollinearity and unstable coefficients, indicating that no robust burden effect was present.

5 Discussion

In this study, I investigated whether antimicrobial-resistance burden (AMR burden) is associated with variation in the mutational spectra of MRSA CC1. Principal component analysis showed that the dominant axis of variation (PC1) primarily reflected differences between studies rather than differences in AMR burden. This pattern was confirmed by linear modelling: in the unadjusted model, AMR burden showed no association with PC1, and the apparent significance emerging after adjusting for study was attributable to confounding. Because studies differed both in their baseline mutational spectra and in the distribution of AMR burden, the study-adjusted model incorrectly attributed between-study differences to burden. Once study–burden interactions were included, the effect disappeared and the model became numerically unstable due to multicollinearity, indicating that no consistent burden effect was present across cohorts.

A similar pattern was observed for the three single-base substitution classes (C>A, C>T and T>G). None of the unadjusted models showed evidence of an association with AMR burden. Although the study-adjusted models suggested positive effects for all three signatures, these associations did not persist in the interaction models. For C>A, the main effect of AMR burden reached nominal significance, but this reflected only the reference study and was contradicted by large, inconsistent interaction terms across the remaining cohorts. For C>T and T>G, the main effects were clearly non-significant. Across all signatures no study-independent relationship between AMR burden and mutational spectra could be established.

Several factors may explain the absence of a robust association. First, MRSA mutational spectra are dominated by endogenous processes, such as spontaneous cytosine deamination (C>T), which may not be strongly influenced by antibiotic exposure [4]. Second, CC1 is not a purely epidemic clone but a lineage also associated with long-term colonisation. Colonising lineages evolve under relatively stable ecological conditions, with limited fluctuations in selective pressures [9, 18]. As a result, their mutational spectra tend to be more conserved, which may explain why no clear association with AMR burden was detectable in this dataset. Third, AMR burden represents an accumulated genomic phenotype rather than a direct measure of current selective pressure. Many resistance determinants may have been acquired outside periods of strong antibiotic exposure, or through horizontal transfer in ecological contexts unrelated to antibiotic stress. Therefore AMR burden may not correlate with ongoing mutational processes.

Overall, my analyses provide no evidence for a consistent, reproducible association between AMR burden and mutational spectra in MRSA, either at the level of global variation (PC1) or at the level of specific mutation classes. Instead, the observed variability appears to be driven primarily by study-specific factors. Future work could benefit from more homogeneous cohorts, longitudinal sampling, or integration of environmental and clinical metadata to better capture potential selective effects of antibiotic exposure.

References

[1] J Andrew N Alexander, Liam J Worrall, Jinhong Hu, Marija Vuckovic, Nidhi Satishkumar, Raymond Poon, Solmaz Sobhanifar, Federico I Rosell, Joshua Jenkins, Daniel Chiang, et al. Structural basis of broad-spectrum β-lactam resistance in staphylococcus aureus. *Nature*, 613(7943):375–382, 2023.

[2] Centers for disease control and prevention. Clinical overview of methicillin-resistant staphylococcus aureus (mrsa) in healthcare settings. Clinical Overwiev, 2025. Accessed: 2025-06-15.

[3] F Coll, EM Harrison, MS Toleman, S Reuter, KE Raven, B Blane, B Palmer, ARM Kappeler, NM Brown, ME Török, et al. Longitudinal genomic surveillance of mrsa in the uk reveals transmission patterns in hospitals and the community. sci transl med 9: eaak9745, 2017.

[4] Bruce K Duncan and Jeffrey H Miller. Mutagenic deamination of cytosine residues in dna. *Nature*, 287(5782):560–561, 1980.

[5] MR Earls, EJ Steinig, S Monecke, JA Samaniego Castruita, A Simbeck, W Schneider-Brachert, et al. Exploring the evolution and epidemiology of european cc1-mrsa-iv: tracking a multidrug-resistant community-associated meticillin-resistant staphylococcus aureus clone. microb genom. 2021; 7 (7). *DOI: https://doi. org/10.1099/mgen. 0.000601. PMID: https://www. ncbi. nlm. nih. gov/pubmed/34223815*, page 000601, 2021.

[6] M Fasnacht and N Polacek. Oxidative stress in bacteria and the central dogma of molecular biology. front mol biosci 8: 671037, 2021.

[7] Joan Geoghegan. The mutational landscape of staphylococcus aureus during colonisation. *Nature Communications*, 2023.

[8] gov.uk. Executive summary third uk one health report. UK One Health Report, 2023. Accessed: 2025-06-15.

[9] Feifei Gu, Weiping He, Dedong Zhu, Qian Zeng, Xinxin Li, Shuzhen Xiao, Yuxing Ni, and Lizhong Han. Genome-wide comparative analysis of cc1 staphylococcus aureus between colonization and infection. *European Journal of Medical Research*, 29(1):474, 2024.

[10] Wei-Ping He, Fei-Fei Gu, Ji Zhang, Xin-Xin Li, Shu-Zhen Xiao, Qian Zeng, Yu-Xing Ni, and Li-Zhong Han. Molecular characteristics and risk factor analysis of staphylococcus aureus colonization put insight into cc1 colonization in three nursing homes in shanghai. *PLoS One*, 16(10):e0253858, 2021.

[11] Teruyo Ito, Yuki Katayama, Kazumi Asada, Namiko Mori, Kanae Tsutsumimoto, Chuntima Tiensasitorn, and Keiichi Hiramatsu. Structural comparison of three types of staphylococcal cassette chromosome mec integrated in the chromosome in methicillin-resistant staphylococcus aureus. *Antimicrobial agents and chemotherapy*, 45(5):1323–1336, 2001.

[12] Sahreena Lakhundi and Kunyan Zhang. Methicillin-resistant staphylococcus aureus: molecular characterization, evolution, and epidemiology. *Clinical microbiology reviews*, 31(4):10–1128, 2018.

[13] Aravind Goud G Patil, Pau Biak Sang, Ashwin Govindan, and Umesh Varshney. Mycobacterium tuberculosis mutt1 (rv2985) and adprase (rv1700) proteins constitute a two-stage mechanism of 8-oxo-dgtp and 8-oxo-gtp detoxification and adenosine to cytidine mutation avoidance. *Journal of Biological Chemistry*, 288(16):11252–11262, 2013.

[14] Torunn Gresdal Rønning, Hege Enger, Jan Egil Afset, and Christina Gabrielsen Ås. Antimicrobial resistance trends of methicillin-resistant staphylococcus aureus in norway from 2008 to 2017. *JAC-Antimicrobial Resistance*, 7(3):dlaf094, 2025.

[15] Christopher Ruis, Aaron Weimann, Gerry Tonkin-Hill, Arun Prasad Pandurangan, Marta Matuszewska, Gemma GR Murray, Roger C Lévesque, Tom L Blundell, R Andres Floto, and Julian Parkhill. Mutational spectra are associated with bacterial niche. *Nature communications*, 14(1):7091, 2023.

[16] Fatemeh Shahkarami, Ahmad Rashki, and Zahra Rashki Ghalehnoo. Microbial susceptibility and plasmid profiles of methicillin-resistant staphylococcus aureus and methicillin-susceptible s. aureus. *Jundishapur journal of microbiology*, 7(7):e16984, 2014.

[17] Martin Vestergaard, Dorte Frees, and Hanne Ingmer. Antibiotic resistance and the mrsa problem. *Microbiology spectrum*, 7(2):10–1128, 2019.

[18] Nicholas Wei Tek Yee, Geoffrey Wallace Coombs, Marc Stegger, Sharmin Baig, Hui-Leen Tan, and Shakeel Mowlaboccus. Genomic epidemiology of clonal complex 1 staphylococcus aureus in remote western australian communities. *International journal of microbiology*, 2025(1):3445177, 2025.